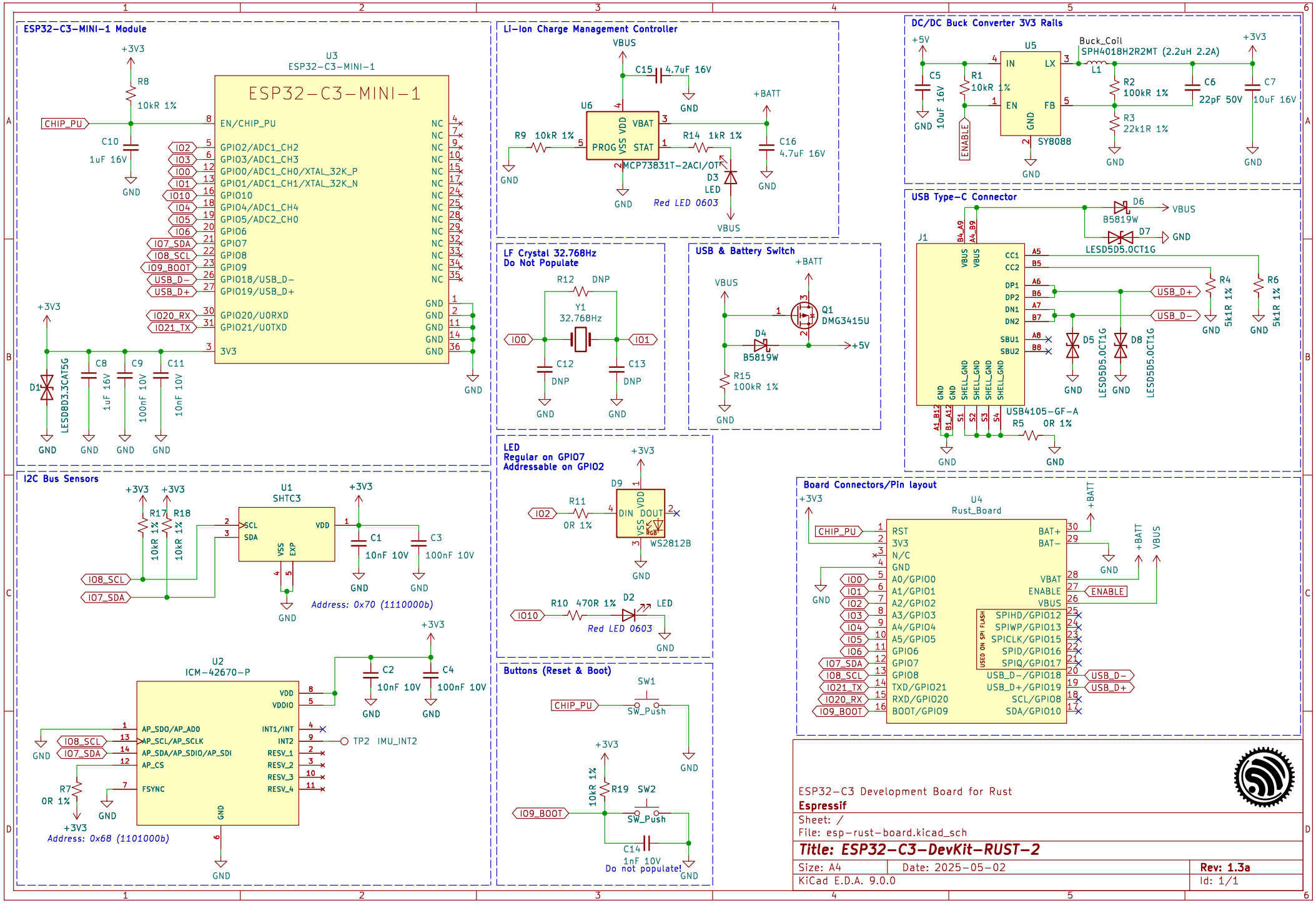

ESP32-C3-MINI-1 Module
U3
ESP32-C3-MINI-1
ESP32-C3-MINI-1
+3V3
R8
10kR 1%
CHIP_PU
C10
1uF 16V
GND
EN/CHIP_PU
IO2
IO3
IO0
IO1
IO10
IO4
IO5
IO6
IO7_SDA
IO8_SCL
IO9_BOOT
USB_D-
USB_D+
IO20_RX
IO21_TX
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO10
GPIO4/ADC1_CH4
GPIO5/ADC2_CH0
GPIO6
GPIO7
GPIO8
GPIO9
GPIO18/USB_D-
GPIO19/USB_D+
GPIO20/U0RXD
GPIO21/U0TXD
3V3
NC
GND
D1
LESD8D3.3CAT5G
C8
1uF 16V
C9
100nF 10V
C11
10nF 10V
Li-Ion Charge Management Controller
VBUS
C15
4.7uF 16V
U6
MCP73831T-2ACI/OT
VDD
VSS
PROG
VBAT
STAT
+BATT
R9 10kR 1%
R14 1kR 1%
C16
4.7uF 16V
D3
LED
Red LED 0603
LF Crystal 32.768Hz
Do Not Populate
R12 DNP
Y1
32.768Hz
C12 DNP
C13 DNP
USB & Battery Switch
Q1
DMG3415U
D4
B5819W
+5V
R15
100kR 1%
DC/DC Buck Converter 3V3 Rails
U5
SY8088
IN
LX
FB
EN
GND
Buck_Coil
SPH4018H2R2MT (2.2uH 2.2A)
L1
R1
10kR 1%
R2
100kR 1%
R3
22k1R 1%
C5
10uF 16V
C6
22pF 50V
C7
10uF 16V
ENABLE
USB Type-C Connector
J1
D6
B5819W
D7
LESD5D5.0CT1G
CC1
CC2
DP1
DP2
DN1
DN2
SBU1
SBU2
SHELL_GND
USB_D+
USB_D-
R4
5k1R 1%
R6
5k1R 1%
D5
LESD5D5.0CT1G
D8
LESD5D5.0CT1G
USB4105-GF-A
R5
0R 1%
I2C Bus Sensors
U1
SHTC3
SCL
SDA
VDD
VSS
EXP
R17
10kR 1%
R18
10kR 1%
C1
10nF 10V
C3
100nF 10V
Address: 0x70 (1110000b)
U2
ICM-42670-P
AP_SDO/AP_ADO
AP_SCL/AP_SCLK
AP_SDA/AP_SDIO/AP_SDI
AP_CS
FSYNC
INT1/INT
INT2
RESV_1
RESV_2
RESV_3
RESV_4
TP2 IMU_INT2
C2
10nF 10V
C4
100nF 10V
R7
0R 1%
Address: 0x68 (1101000b)
LED
Regular on GPIO7
Addressable on GPIO2
D9
WS2812B
DIN
DOUT
R11
0R 1%
R10 470R 1%
D2
LED
Red LED 0603
Buttons (Reset & Boot)
SW1
SW_Push
R19
10kR 1%
SW2
C14
1nF 10V
Do not populate!
Board Connectors/Pin layout
U4
Rust_Board
RST
3V3
N/C
GND
A0/GPIO0
A1/GPIO1
A2/GPIO2
A3/GPIO3
A4/GPIO4
A5/GPIO5
GPIO6
GPIO7
GPIO8
TXD/GPIO21
RXD/GPIO20
BOOT/GPIO9
BAT+
BAT-
VBAT
ENABLE
VBUS
SPIHD/GPIO12
SPIWP/GPIO13
SPICLK/GPIO15
SPID/GPIO16
SPIQ/GPIO17
USB_D-/GPIO18
USB_D+/GPIO19
SCL/GPIO8
SDA/GPIO10
USED ON SPI FLASH
ESP32-C3 Development Board for Rust
Espressif
Sheet: /
File: esp-rust-board.kicad_sch
Title: ESP32-C3-DevKit-RUST-2
Size: A4
Date: 2025-05-02
Rev: 1.3a
KiCad E.D.A. 9.0.0
Id: 1/1